

Emergency Department utilization with age, race, and access across Kansas

Presented by : Janice Leung, Yuliia Veres, Minal Rehan, Aarya Gokhale

Problem Statement



Overview

Healthcare utilization varies across populations due to differences in access, cost, and social factors.

Research Question

How can we predict ED visits based on the interaction between age and race?

Goal

Identify patterns in emergency department use to better understand population-level differences in health-care utilization.

Our Data Setup

Data Preparation

Data Integration

Linked patient +
encounter data using
DurableKey (patients) +
PatientDurableKey
(encounters)

Outcome Variable

Created patient-level
outcome:
Had ED Visit (Yes (1) /No(0))
Missing encounters → treated
as no ED visit

Data Filtering

Filtered:
Living patients
Non-missing race &
age

Methods Used

Logistic Regression

Outcome: ED visit (binary)

Predictors: Age & Race

Age × Race interaction

ED visits rare (~3%)

→ Used a stratified split to create representative data splits by preserving outcome proportions in the training and test sets.

Results: Emergency Medicine, Radiology, and Neonatology are most visited departments.

Visits × City

→ Used to determine domination of in-demand medical care departments

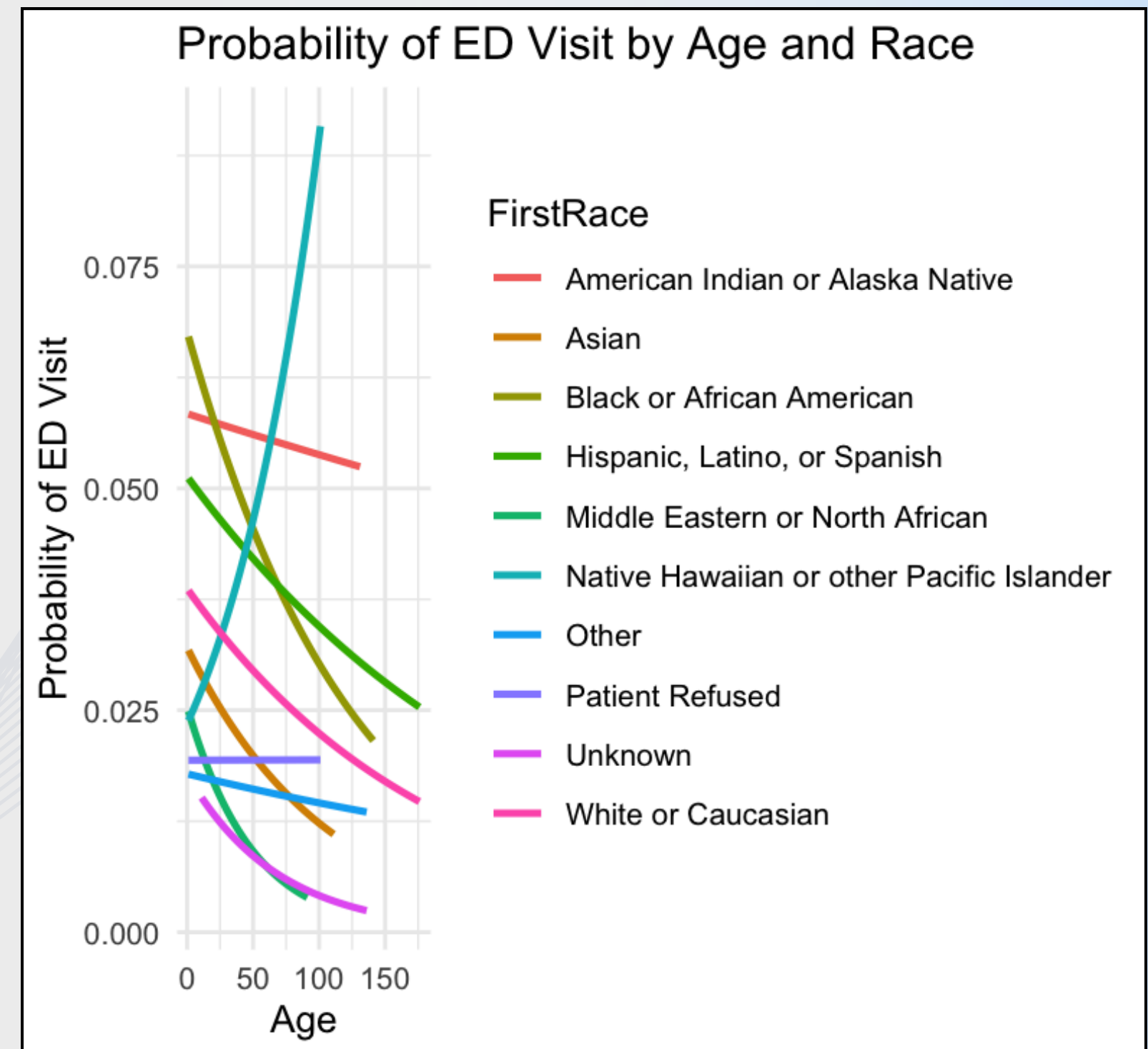
→ Most visited department varies per city.

Data Filtering and Grouping

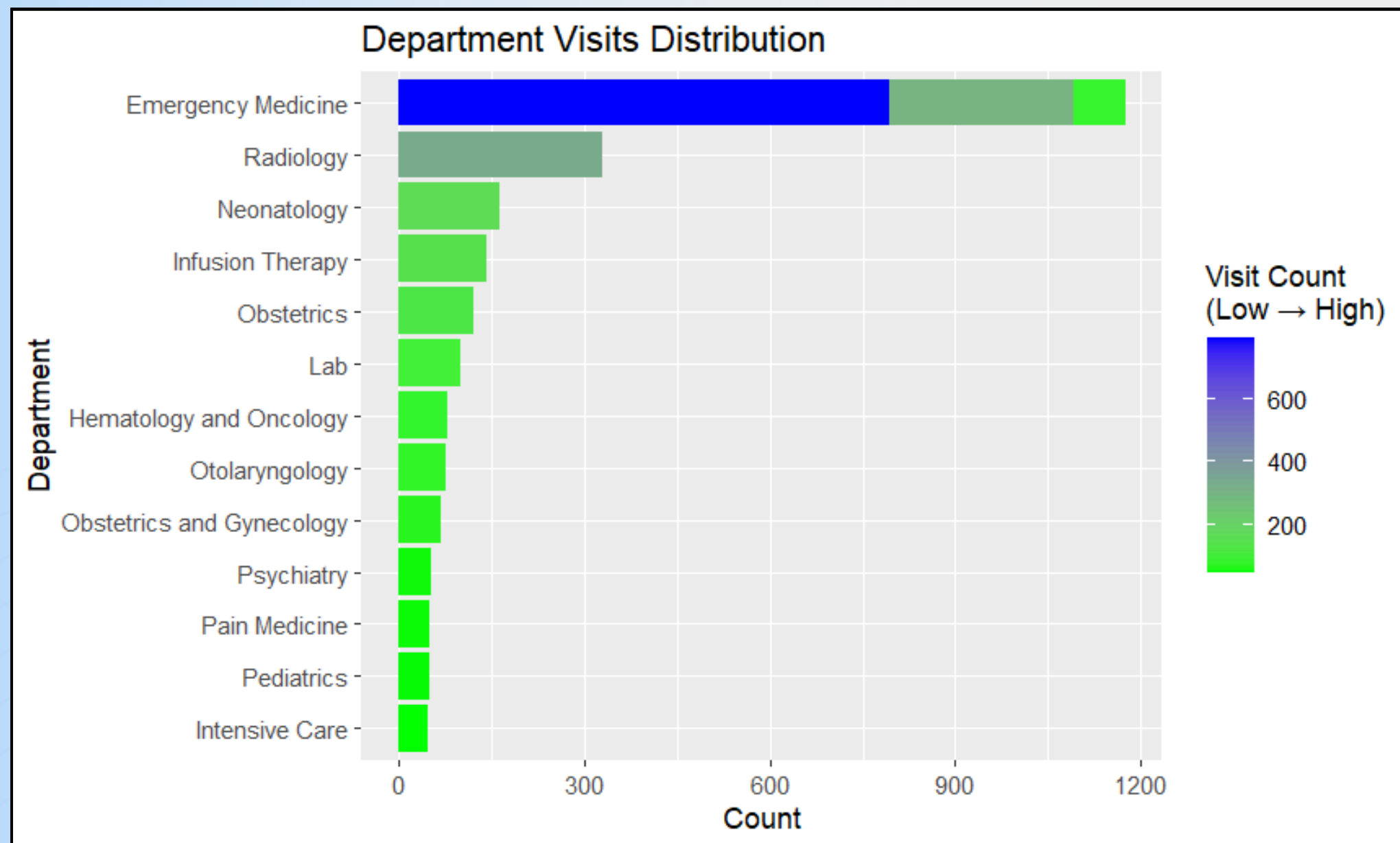
Key insights & Results

ED Visit Probability by Age and Race

- ED visits are **rare** across all ages
- Age alone → **weak** effect
- Patterns **vary slightly** across groups



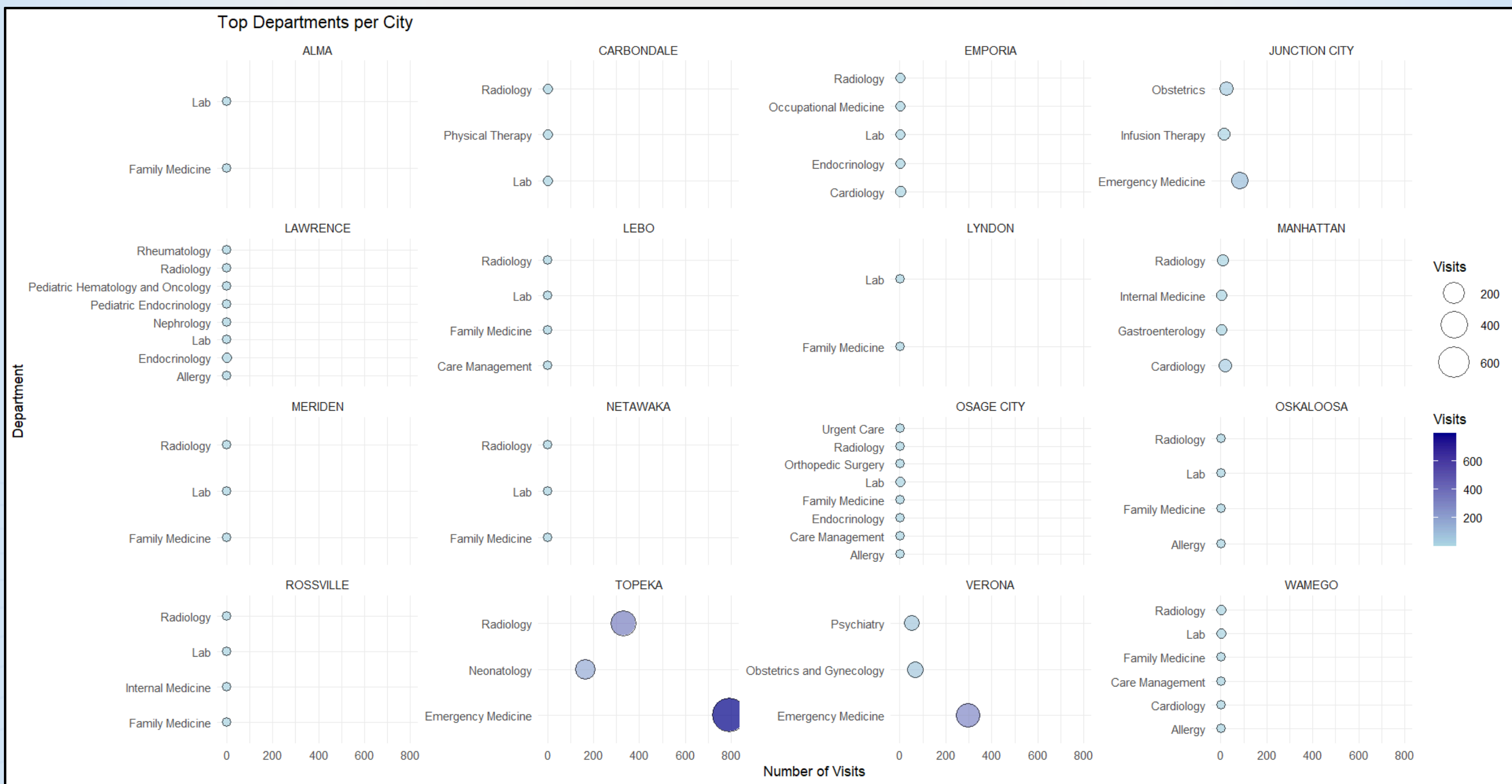
Department Analysis



Where Are Visits Concentrated?

- **Emergency Medicine** and **Radiology** departments have most visits
- All other departments have significant lower visits
- **Visit distribution per department is highly uneven**

Most visited departments across cities



Summary & Key Takeaways

Emergency Department

- Visits are low overall population (~3%)

Age & Race

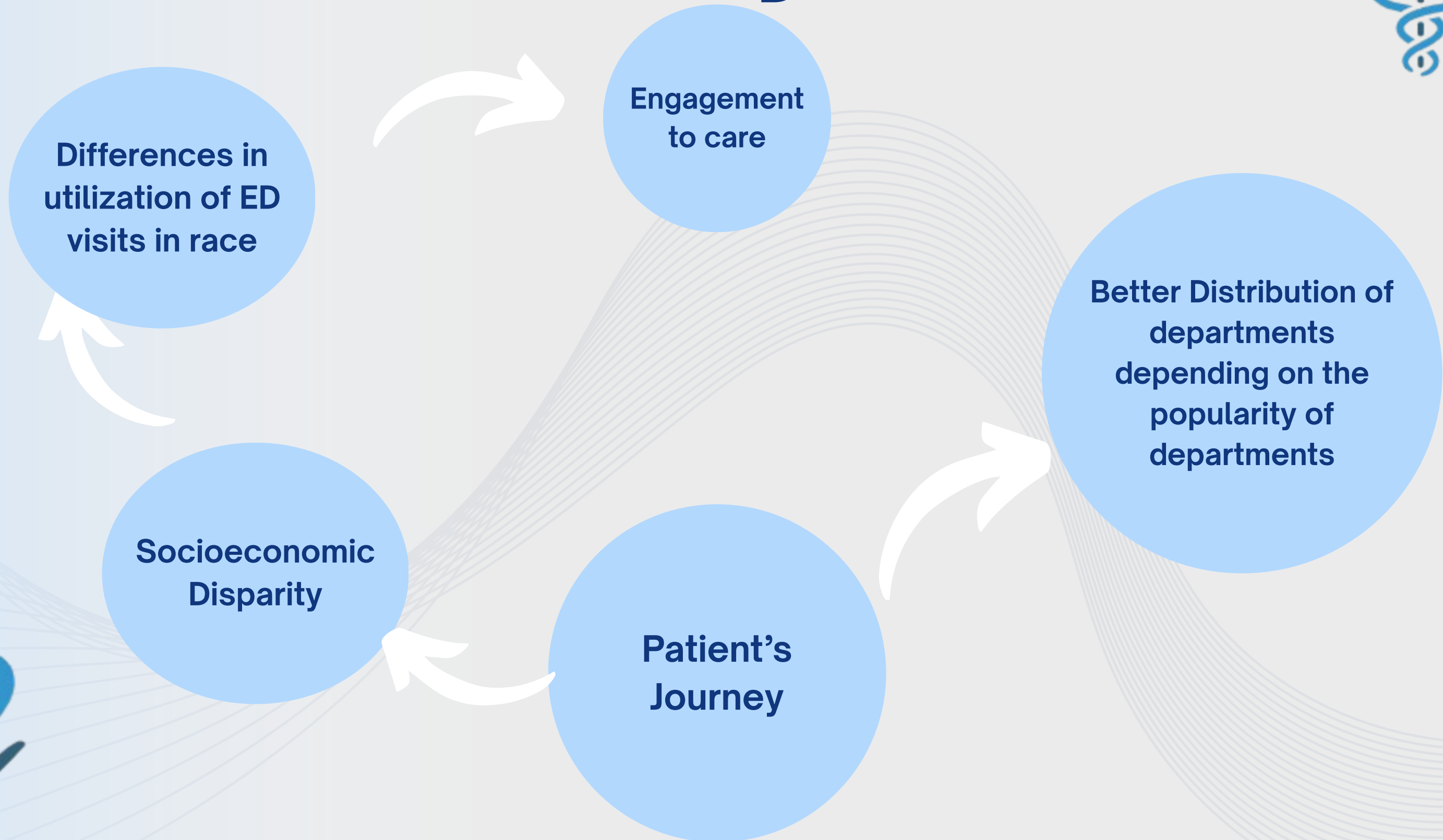
- Age alone → not a strong driver
- Modest differences exist across racial groups

Revealing results

- Dataset size is large adding complexity and the need for further research



Patient's Journey



Differences in
utilization of ED
visits in race

Engagement
to care

Better Distribution of
departments
depending on the
popularity of
departments

Patient's
Journey

Socioeconomic
Disparity

THANK YOU!

Q & A

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